

FACTORSPLAT: APPEARANCE-CONTROLLABLE GAUSSIAN PROXIES FOR MEDICAL VOLUME RENDERING

Zhongpai Gao^{1*}, Benjamin Planche¹, Meng Zheng¹, Anwesa Choudhuri¹,
Terrence Chen², Ziyang Wu¹

¹United Imaging Intelligence, Boston MA, USA
{first.last}@uii-ai.com

²United Imaging Healthcare Co.

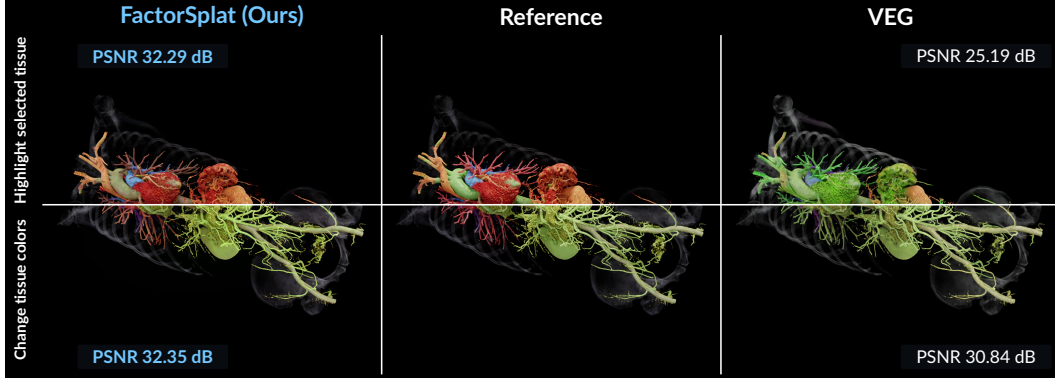


Figure 1: **Change appearance without retraining.** Two vascular TF edits: highlight selected tissue (top) and change tissue colors (bottom). Full-frame PSNR.

ABSTRACT

Transfer functions (TFs) control color and visibility in medical volume rendering, but image-trained Gaussian proxies typically bake one transfer function into their appearance. We present FactorSplat, a per-scene N-dimensional Gaussian splatting (N-DGS) proxy that accepts region-specific intensity-to-RGBA curves at inference. A local lookup applies the authored color and opacity change, while a shared functional encoder and low-rank per-Gaussian factors learn the residual appearance response. Geometry and directional appearance remain shared across presets, with visibility control and TF-aware pruning preserving the ability to hide and reveal structures. On seven CT and MR scans, FactorSplat improves mean PSNR and changed-region error over region-aware VEG across validation, interpolation, unseen composition, and out-of-distribution (OOD) edits. Across these four splits, seven-scan mean PSNR gains over VEG range from 1.10 to 1.52 dB. One checkpoint per scan supports unseen edits without retraining. At 1600², the cached fast renderer averages 524 FPS with 1.17 ms TF switches. The project page is: <https://gaozhongpai.github.io/FactorSplat/>.

1 INTRODUCTION

Transfer functions (TFs) control the appearance of volume renderings by mapping measured intensity and derived attributes to color and opacity. They can suppress one material, expose another, or separate structures that overlap in a one-dimensional histogram (Kniss et al., 2002). In the clinical renderer considered here, each segmented region has an authored intensity-to-RGBA curve (HU for CT). Segmentation selects the appropriate curve, whose resulting color and opacity govern volumetric transport. A preset change can therefore alter both which anatomy contributes to the image and how the visible anatomy occludes, shades, and transmits light in cinematic rendering (Dappa et al., 2016).

*Corresponding author

Image-trained Gaussian proxies offer an efficient representation of such renderings. A proxy optimized from high-quality volume renderings is compact and rasterizes in real time (Niedermayr et al., 2024; Gao et al., 2026). These fixed-appearance proxies encode the training appearance in per-Gaussian spherical-harmonic coefficients and opacities, so the TF is no longer an input. Supporting another preset then requires rebuilding or replacing the proxy. Moreover, training on one opaque preset may omit anatomy that a later preset should reveal.

Making the TF an inference-time control poses three coupled problems. First, the condition is a *function*, not a categorical preset identifier: a model should respond to a new sampled curve without allocating new parameters. Second, the response is spatially heterogeneous. The same TF edit should affect a Gaussian according to the region curves and intensities it covers. Third, training and densification must preserve *coverage* across the condition family so that structures revealed by any supported preset remain represented. Because a TF changes optical appearance rather than anatomy, geometry should remain invariant to this control.

We address these challenges with FactorSplat, a per-scene Gaussian proxy trained across TFs. Gaussian centers, covariances, and the N-DGS view response are shared. Local region–intensity samples drive two complementary appearance branches: a physical lookup applies the TF difference directly, while a shared pointwise encoder and low-rank per-Gaussian factors model the cinematic renderer’s residual response. The lookup anchors the prediction to the authored edit; the learned branch corrects its image-space effect. Both branches modify only the view-independent (DC) color and opacity; directional SH coefficients remain TF-independent. Centering the branches on an authored base TF gives exact identity at that preset, and pruning by maximum opacity over the training bank protects primitives needed by different training TFs.

Our main contributions are:

- **Representation:** FactorSplat conditions an N-DGS proxy on region-specific intensity-to-RGBA curves, rendering unseen TFs with shared geometry and no refitting.
- **Appearance model:** A hybrid model combines an interpretable local TF lookup, a low-rank DC-color/opacity residual, analytic visibility control, and TF-aware coverage preservation.
- **Evaluation:** Across seven scans, one checkpoint per scan improves mean PSNR and edit fidelity over region-aware VEG on unseen TFs.

2 RELATED WORK

Transfer functions and learned visualization synthesis. Multidimensional TFs use intensity, gradients, or other attributes to classify overlapping volume structures and assign optical properties interactively (Kniss et al., 2002). A complementary line of work helps users *choose* that mapping: differentiable direct volume rendering optimizes TF parameters against an image objective (Weiss & Westermann, 2022), and recent semantic systems derive mappings from selected structures or text prompts (Engel et al., 2024; Jeong et al., 2024). FactorSplat reproduces the rendering of an authored TF with a compact proxy.

Image-space visualization surrogates condition neural generators on viewpoint and rendering parameters. Berger et al. (2019) encode color and opacity TFs into a view-invariant latent space, while DNN-VolVis (Hong et al., 2019) predicts volume-rendered images for interactive exploration. These models can learn the output of either local or global-illumination renderers, but an image generator has no explicit 3D primitive representation. ViSNeRF introduces a factorized radiance field for interpolation over TFs, isovalues, time, and simulation parameters (Yao et al., 2025); it retains ray-based field rendering rather than an explicit rasterizable proxy. Our functional encoder likewise keeps rendering controls explicit, but its output modulates a per-scene Gaussian field.

Gaussian proxies for volume rendering. 3D Gaussian proxies trained on multi-view volume renderings enable compact, real-time viewing of cinematic anatomy (Niedermayr et al., 2024); Render-FM amortizes proxy construction across scans by predicting an N-DGS representation feed-forward (Gao et al., 2026). Both target a fixed rendering appearance. iVR-GS instead trains editable basic models for disjoint intensity ranges and composes them to control color, opacity, and Blinn–Phong lighting (Tang et al., 2025). Its cost and representational size grow with the number of

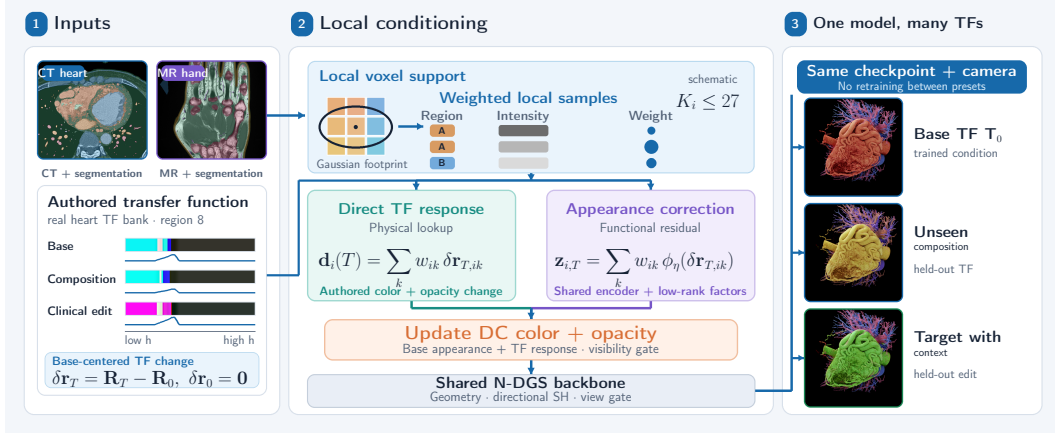


Figure 2: **Method overview.** FactorSplat combines a physical lookup and low-rank residual over local material–intensity support to condition DC color and opacity.

basic models. ECoNGS shares parameters and compresses anchors across basic scenes, reducing redundancy while retaining the composition model (Tang & Wang, 2026).

VEG is the closest transfer-function-agnostic Gaussian representation (Dyken et al., 2026). It initializes Gaussians from structured or unstructured volume data, stores scalar values rather than baked color and opacity, and trains under a set of opacity maps that covers the scalar range. At inference, it directly applies unseen one-dimensional color and opacity mappings under an emission–absorption render with Blinn–Phong shading. This factorization directly links Gaussian appearance to the scalar field. Our setting instead uses region-specific intensity TFs and cinematic-rendering supervision. We combine direct lookup with an image-trained N-DGS proxy whose residual learns the TF-dependent appearance response. Section 4 describes our region-aware VEG extension.

Conditioned explicit Gaussian fields. N-dimensional Gaussian splatting has conditioned primitives on viewing direction and time (Gao et al., 2025a;b; Liu et al., 2026). FactorSplat uses the opacity-only N-DGS formulation as its view-dependent backbone and adds a separate TF axis only to appearance. Concurrent scientific visualization systems also condition Gaussian attributes. GS-Surrogate learns sequential simulation- and visualization-parameter deformations and disables geometry heads for its opacity-only TF experiment (Li et al., 2026); ParticleGS maps global particle radius and opacity controls to Gaussian scale and opacity (Jiang et al., 2026). These methods condition Gaussian fields on low-dimensional controls. DeepONet (Lu et al., 2021) encodes input functions for a separable output representation. FactorSplat instead grounds function-valued TF inputs in each Gaussian’s local support, with per-primitive factors mapping local codes to appearance offsets rather than generating full weights with a hypernetwork (Ha et al., 2017).

3 METHOD

Figure 2 separates shared scene structure from TF-dependent appearance. Local material descriptors connect authored curves to Gaussian color and opacity through a direct lookup and a learned residual. TF changes update these attributes once; camera navigation reuses them.

3.1 PROBLEM FORMULATION AND N-DGS BACKBONE

A scene contains N Gaussian primitives rasterized by EWA splatting and front-to-back alpha compositing (Zwicker et al., 2001; Kerbl et al., 2023). For primitive i , $\mu_i \in \mathbb{R}^3$, $\Sigma_i \in \mathbb{R}^{3 \times 3}$, and $o_i \in \mathbb{R}$ denote its center, covariance, and opacity logit; its base opacity is $\sigma(o_i)$ for logistic sigmoid σ . Every N-DGS-based model in our comparison uses degree-one spherical harmonics: $Q_b = 4$ basis functions, one DC and three directional. Let $\mathbf{C}_i^{\text{base}} \in \mathbb{R}^{Q_b \times 3}$ collect their RGB coefficients and $Y_q(\omega_i(\pi))$ denote basis q evaluated at the viewing direction from primitive i to camera π . Before the usual rasterizer activation, the shared backbone color is $\sum_{q=0}^{Q_b-1} Y_q(\omega_i(\pi)) \mathbf{C}_{i,q}^{\text{base}}$, with $q = 0$ denoting DC.

We use an opacity-only N-DGS backbone (Gao et al., 2025a), with directly parameterized view-dependent opacity and no view-dependent position shifts. Let π denote a camera and $s_i^{\text{view}}(\pi) \in (0, 1]$ the TF-independent opacity attenuation produced by N-DGS for primitive i . The backbone opacity passed to the rasterizer is $\sigma(o_i)s_i^{\text{view}}(\pi)$. FactorSplat leaves this gate unchanged and keeps all geometry shared across TFs: $\mu_i(\pi, T) = \mu_i$ and $\Sigma_i(\pi, T) = \Sigma_i$.

A preset contains one intensity-to-RGBA *transfer curve* for every segmented renderer region. Writing m for the region/material channel and h for the calibrated scalar intensity,

$$\mathbf{R}_T(m, h) = (\rho_T(m, h), a_T(m, h)) \in [0, 1]^3 \times [0, 1]. \quad (1)$$

Here h is calibrated HU for CT and the renderer’s normalized scalar for MR; $m \in \{0, \dots, M-1\}$ indexes one of M region curves, ρ_T is RGB, and a_T is transfer opacity. The index m routes a voxel to its authored curve; the encoder consumes the curve’s RGBA values rather than the region identifier. Exporting every preset on the common grid $\{h_s\}_{s=1}^S$ gives the sampled input $\mathbf{R}_T \in [0, 1]^{M \times S \times 4}$; binary region-visibility flags are denoted separately by v_T below. We designate one authored preset T_0 as the base condition.

Given reference images for a training bank $\mathcal{T}_{\text{train}}$, we jointly learn one proxy that renders unseen $T \notin \mathcal{T}_{\text{train}}$ without optimization or per-preset parameters.

3.2 LOCAL TRANSFER-FUNCTION CONDITIONING

A Gaussian fitted to cinematic images combines material response, shading, and visibility. We therefore separate the authored TF change from its effect on the fitted proxy. The physical lookup supplies a local color and opacity offset; the learned residual corrects the primitive’s DC color and opacity for the same edit. Geometry and directional appearance are shared across TFs.

Local material–intensity descriptor and physical lookup. Let $\mathbf{R}_T^{(B)} \in [0, 1]^{M \times B \times 4}$ denote the raw TF curves averaged onto $B = 64$ intensity bins. At initialization, we sample the 3^3 voxel window around primitive i . Its k th valid sample supplies a region channel $m_{ik} \in \{0, \dots, M-1\}$ and an intensity-bin index $b_{ik} \in \{0, \dots, B-1\}$; $K_i \leq K = 27$ is the number of valid samples. Normalized nonnegative weights w_{ik} satisfy $\sum_{k=1}^{K_i} w_{ik} = 1$ and define the empirical joint mass $p_i(m, b) = \sum_k w_{ik} \mathbf{1}[(m_{ik}, b_{ik}) = (m, b)]$. Both conditional branches read the same quantity, the raw, unpremultiplied RGBA change that the preset applies to one region–intensity entry,

$$\delta \mathbf{r}_T(m, b) = \mathbf{R}_T^{(B)}(m, b) - \mathbf{R}_{T_0}^{(B)}(m, b) \in \mathbb{R}^4. \quad (2)$$

Keeping the channels unpremultiplied separates color and opacity offsets. The physical *local lookup* is the weighted mean of this change over the primitive’s own samples,

$$\mathbf{d}_i(T) = \sum_{k=1}^{K_i} w_{ik} \delta \mathbf{r}_T(m_{ik}, b_{ik}). \quad (3)$$

Here $\mathbf{d}_i = (\mathbf{d}_i^c, d_i^a) \in \mathbb{R}^3 \times \mathbb{R}$ and $\mathbf{d}_i = \mathbf{0}$ when $K_i = 0$. We encode each pair as $u_{ik} = m_{ik}B + b_{ik}$. The region index only selects the corresponding TF curve: permuting the curve axis and identifiers together leaves the lookup unchanged. Material associations persist during training (Section 3.4).

Local functional encoder. The learned branch embeds each RGBA change before weighted, permutation-invariant pooling, following the set-encoding principle of Deep Sets (Zaheer et al., 2017). A bias-free pointwise MLP $\phi_\eta(\mathbf{v}) = W_2 \text{ReLU}(W_1 \mathbf{v})$, with $W_1 \in \mathbb{R}^{H \times 4}$ and $W_2 \in \mathbb{R}^{r \times H}$, maps each RGBA difference to a shared response basis. Primitive i pools its local codes:

$$\mathbf{z}_{i,T} = \sum_{k=1}^{K_i} w_{ik} \phi_\eta(\delta \mathbf{r}_T(m_{ik}, b_{ik})) \in \mathbb{R}^r, \quad (4)$$

with $\mathbf{z}_{i,T} = \mathbf{0}$ when $K_i = 0$. The MLP is evaluated once over the $M \times B$ table at a TF switch; packed gathers then produce all primitive codes. Each code depends on the primitive’s own support, allowing spatially different responses to the same preset. The shared RGBA encoder accepts unseen TFs without preset-specific parameters. Centering and bias-free layers ensure $\mathbf{z}_{i,T_0} = \mathbf{0}$ exactly. Encoding before pooling retains responses to individual local edits that could cancel if their RGBA differences were averaged first. For example, changing a bone curve contributes only to codes whose descriptors include bone, while soft-tissue edits act through their own support.

3.3 FACTORIZED APPEARANCE RESPONSE

Low-rank per-Gaussian factors. Each primitive carries color factors $\mathbf{A}_i^c \in \mathbb{R}^{3 \times r}$ and opacity factors $\mathbf{a}_i^\alpha \in \mathbb{R}^r$. They translate the local code into a DC-SH residual $\mathbf{r}_i^c \in \mathbb{R}^3$ and an opacity-logit residual $r_i^\alpha \in \mathbb{R}$. Define the authored alpha-change indicator

$$q^\alpha(T) = \mathbf{1} \left[\left\| \mathbf{R}_T^{(B)}[:, :, \alpha] - \mathbf{R}_{T_0}^{(B)}[:, :, \alpha] \right\|_\infty > 0 \right]. \quad (5)$$

The factor responses are

$$\mathbf{r}_i^c(T) = \lambda_R^c \tanh(\mathbf{A}_i^c \mathbf{z}_{i,T}), \quad r_i^\alpha(T) = q^\alpha(T) \lambda_R^\alpha (\mathbf{a}_i^\alpha)^\top \mathbf{z}_{i,T}. \quad (6)$$

Here $\lambda_R^c, \lambda_R^\alpha > 0$ are fixed output scales and $\tanh$ acts elementwise. For RGB-only edits, $q^\alpha = 0$ keeps opacity unchanged; authored alpha edits enable the learned opacity response. Stacking all color and opacity factors yields a $4N \times r$ matrix. The latent dimension r controls the size of the shared response basis before the output nonlinearity, independently of the number of TFs. The factors, shared encoder, and base appearance are jointly optimized.

Complete appearance model. Let $b_L, b_R \in \{0, 1\}$ enable the lookup and residual branches, respectively, and let $\mathbf{g}^c \in \mathbb{R}^3$ and $g^\alpha \in \mathbb{R}$ be the learned global lookup gains. With fixed positive channel scales $\lambda_L^c, \lambda_L^\alpha, \lambda_R^c, \lambda_R^\alpha$, we combine both branches as

$$\mathbf{C}_i(T) = \mathbf{C}_i^{\text{base}} + \mathbf{e}_0 \left[b_L \frac{\lambda_L^c}{C_0} (\mathbf{g}^c \odot \mathbf{d}_i^c(T)) + b_R \mathbf{r}_i^c(T) \right]^\top, \quad (7)$$

$$\bar{\alpha}_i(T) = \sigma(o_i + b_L \lambda_L^\alpha g^\alpha d_i^\alpha(T) + b_R r_i^\alpha(T)), \quad (8)$$

$$\alpha_i(\pi, T) = \gamma_i(T) \bar{\alpha}_i(T) s_i^{\text{view}}(\pi), \quad (9)$$

where $\mathbf{e}_0 = (1, 0, 0, 0)^\top \in \mathbb{R}^{Q_b}$ selects the DC row, $C_0 = 1/(2\sqrt{\pi})$ converts RGB offsets to DC-SH coefficients, $\odot$ denotes componentwise multiplication, and $\bar{\alpha}_i(T)$ is the TF-conditioned opacity before the shared N-DGS view gate. The settings $(b_L, b_R) = (1, 0), (0, 1), (1, 1)$ define the *lookup*, *residual*, and *hybrid* variants, respectively. FactorSplat uses $(b_L, b_R) = (1, 1)$. Both offsets vanish at T_0 , preserving the shared base appearance. The gate $\gamma_i(T)$ applies binary region removal as defined below. At inference, conditioned SH and opacity are cached once per TF switch; camera motion then uses the unchanged N-DGS rasterizer.

Analytic visibility gate. Authored binary region visibility is applied analytically. Let $v_T(m) \in \{0, 1\}$ be the renderer’s authored visibility flag for region m , and define the indicator of removal relative to the base preset as $r_T(m) = v_{T_0}(m)[1 - v_T(m)]$. The removed material mass of primitive i is

$$\eta_i(T) = \sum_{m,b} p_i(m, b) r_T(m), \quad \gamma_i(T) = \mathbf{1}[\eta_i(T) < \tau], \quad \tau = \frac{1}{2}. \quad (10)$$

Primitives without descriptor support use $\gamma_i = 1$. Equation 9 multiplies this gate *after* the sigmoid, so a gated primitive has exactly zero opacity. For mixed support, removal occurs when the combined mass of removed regions reaches one half. Defining removal relative to T_0 preserves the base render. This parameter-free gate handles binary visibility; the conditional branches handle graded appearance.

3.4 INITIALIZATION AND TRAINING

Conditioning appearance is useful only if the shared geometry represents structures that later edits reveal. We initialize Gaussians on a preset-neutral grid covering the union of segmented structures, including those transparent under T_0 . We optimize the standard photometric loss through the conditioned renderer over a mixed-preset batch stream,

$$\hat{\mathbf{I}} = \mathcal{R}(\mathcal{G}; \pi, T), \quad \mathcal{L} = \mathbb{E}_{(\mathbf{I}, \pi, T)} \left[(1 - \lambda_s) \mathcal{L}_1(\hat{\mathbf{I}}, \mathbf{I}) + \lambda_s (1 - \text{SSIM}(\hat{\mathbf{I}}, \mathbf{I})) \right], \quad (11)$$

where $\mathbf{I}$ is the reference RGB image for camera π and preset T , $\mathcal{G}$ denotes all Gaussian and conditioning parameters, $\mathcal{R}$ is the unmodified tile rasterizer, and $\mathcal{L}_1$ is mean absolute RGB error. SSIM measures structural similarity (Wang et al., 2004); we use $\lambda_s = 0.2$. Gradients reach the factors, encoder, and base parameters through the same compositing as at test time. Each training frame

supplies its sampled TF. Anchor cameras shared across presets provide paired appearance changes, while per-preset cameras add view coverage.

Coverage-aware pruning. A primitive needed by one preset may be nearly transparent under another. We therefore replace the base-opacity pruning score $\sigma(o_i)$ with $\kappa_i = \max_{T \in \mathcal{T}_{\text{train}}} \bar{\alpha}_i(T)$. A primitive survives this opacity test if its opacity exceeds the pruning threshold under any training preset. The score excludes camera-dependent attenuation s_i^{view} ; standard geometric and screen-size pruning criteria remain active.

Persistent material descriptors. The samples (m_{ik}, b_{ik}) are attached when the preset-neutral grid is initialized. Their weights measure the initial Gaussian support at each sampled voxel,

$$w_{ik} \propto \exp\left(-\frac{1}{2}(\mathbf{x}_{ik} - \boldsymbol{\mu}_i)^\top \boldsymbol{\Sigma}_i^{-1}(\mathbf{x}_{ik} - \boldsymbol{\mu}_i)\right), \quad \sum_k w_{ik} = 1, \quad (12)$$

where $(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$ are the initial center and covariance and $\mathbf{x}_{ik}$ is a sampled voxel center. Equation 12 makes the descriptor scale-aware: a sub-voxel primitive concentrates on its own material, whereas a larger primitive integrates over its footprint.

Identifiers and weights remain fixed as each primitive’s geometry is optimized. Clone and split children inherit descriptors and factor rows from the nearest pre-existing primitive; pruning applies one mask to all attributes. TF changes therefore use persistent material support rather than reassigning regions from the optimized centers.

4 EXPERIMENTS

4.1 PROTOCOL

Data and training. We evaluate seven scans: five CT (heart, vascular, intestine, lower, and knee joint) and two MR (nose and hand), with 3–17 authored region curves per scan. Each scan uses 41 conditions: 24 training presets, 4 validation presets, 4 interpolation tests, 4 unseen compositions of individually observed operations, and 5 OOD tests. Interpolation varies edits within the training range, while composition combines familiar operations in unseen ways. OOD includes two functional stress tests (OOD-F: grayscale and opacity sharpening) and three anatomy-directed edits (OOD-C: target isolation, occluder suppression, and target emphasis with translucent context). These test new operations rather than new scans; all models are fitted per scan. Each training preset has 48 views (12 shared anchors and 36 preset-specific cameras), while all presets share 40 held-out cameras. We generate ground-truth RGB images at 1600^2 using cinematic volume path tracing of the original CT or MR volumes and segmentation masks. Each preset applies region-specific intensity-to-color and opacity curves. Matched-view renders keep cameras, lighting, and sampling settings fixed across TFs. All methods use the same held-out cameras and 30,000 iterations; conditional models share the 24 training presets. N-DGS-based methods use volume-derived point initializations. VEG samples its native random initialization within the source bounds and with the source point count. FactorSplat uses rank $r = 8$, 64 hidden units, and Mip-Splatting’s 3D filter (Yu et al., 2024).

TF-aware metrics. We report full-frame PSNR for reconstruction quality and a paired-view metric for edit fidelity. The latter isolates the predicted appearance change from the base reconstruction, including edits confined to small regions. Let $\mathbf{D}^{\text{gt}} = \mathbf{I}^{\text{gt}}(T) - \mathbf{I}^{\text{gt}}(T_0)$ and define $\mathbf{D}^{\text{pred}}$ analogously at the same camera. The changed region is $M_T = \{x : \|\mathbf{D}^{\text{gt}}(x)\|_\infty > 0.04\}$. For $\Omega \subseteq \mathcal{P}$, the transfer-function delta error is

$$\Delta_{\text{TF}}(T; \Omega) = \frac{1}{3|\Omega|} \sum_{x \in \Omega} \|\mathbf{D}^{\text{pred}}(x) - \mathbf{D}^{\text{gt}}(x)\|_1. \quad (13)$$

Here $\mathcal{P}$ is the image pixel domain and RGB values lie in $[0, 1]$. We report changed-region error with $\Omega = M_T$. Unchanged-region leak instead averages $\|\mathbf{D}^{\text{pred}}(x)\|_1$ over $\mathcal{P} \setminus M_T$. For an unconditioned proxy, $\Delta_{\text{TF}}(T; M_T)$ equals the magnitude of the reference edit and thus defines the identity floor.

Comparisons. The *N-DGS identity floor* uses FactorSplat’s pooled training images without TF input. Each *N-DGS specialist* fits 48 views of one target TF, including TFs withheld from conditional-model training. FactorSplat and VEG pool all 24 training presets. Specialists thus receive target-TF supervision but less camera coverage; they are preset-specific references, not quality upper bounds.

Table 1: **CT/MR quality and efficiency.** Identity floor: no TF input; specialists: per-target-TF training. B200 FPS at 1600²; OOD: five edits. Per-scan results: Table 3; timing: Section 4.4.

		$\Delta_{TF} \downarrow (\times 10^{-2})$				PSNR $\uparrow$				FPS $\uparrow$	Train (min) $\downarrow$
Model		val	interp	comp	OOD	val	interp	comp	OOD		
CT (5 scans)	N-DGS (identity floor)	3.60	6.49	8.96	17.74	31.11	27.88	25.50	21.11	851	6.9
	N-DGS (specialist)			—		30.70	30.82	31.09	31.94	866	92.8
	VEG	1.40	2.55	3.05	9.15	31.80	30.90	30.64	25.37	134	44.3
	VEG (native-quantized)	1.41	2.56	3.06	9.17	31.65	30.78	30.54	25.32	136	44.3
	FactorSplat (Ours)	1.36	2.17	2.88	8.09	33.08	32.29	31.33	26.58	508	10.5
MR (2 scans)	N-DGS (identity floor)	2.88	4.25	5.48	11.09	36.58	35.54	31.17	27.86	925	6.0
	N-DGS (specialist)			—		37.03	37.04	37.07	38.00	930	76.4
	VEG	1.19	1.89	1.59	5.16	38.82	38.52	37.82	31.24	169	42.6
	VEG (native-quantized)	1.19	1.90	1.59	5.15	38.67	38.39	37.70	31.20	171	42.6
	FactorSplat (Ours)	1.09	1.62	1.28	4.65	40.68	40.36	39.95	33.13	562	8.3

We use the official VEG implementation (Dyken et al., 2026), extending its scalar lookup with a fixed region per Gaussian to select the corresponding authored curve. VEG retains its native shading, optimizer, density control, and 100-bin intensity lookup. In figures and tables, VEG denotes the floating-point model; its compressed export is labeled native-quantized. Appendix B specifies the region-aware extension.

4.2 RECONSTRUCTION QUALITY AND EDIT FIDELITY

Table 1 summarizes the comparison with the N-DGS references and region-aware VEG as separate CT and MR means, weighting scans equally within each modality. Appendix Table 3 reports each scan and the seven-scan mean. FactorSplat improves PSNR over floating-point VEG in 24 of 28 scene-split comparisons, with mean gains of 1.45/1.52/1.10/1.41 dB on validation, interpolation, composition, and OOD. Changed-region error is lower in 19 of 28 comparisons, with lower means on every split for both CT and MR. The two metrics capture distinct aspects of the result: on heart, FactorSplat has higher PSNR on validation, interpolation, and composition, while VEG more accurately reproduces the change from the base image.

Shared versus preset-specific models. FactorSplat has higher PSNR than specialists on validation and interpolation for all seven scans, by up to 4.0 and 3.6 dB. On composition it leads on four scans, ties on one, and trails on two. Its shared geometry receives broader camera coverage across presets than each 48-view specialist. This may contribute to the observed advantage, which does not isolate the effect of conditioning. Specialists retain a 5.2 dB mean advantage on OOD, highlighting the value of direct target-TF fitting. VEG also leads on heart OOD and several kneejoint splits; new edit operations remain harder than interpolation.

Figure 3 shows matched held-out CT and MR views, including differences in tissue color and visibility under unseen edits. Figure 4 provides additional examples.

4.3 ABLATION STUDIES

Table 2 isolates the physical lookup, learned residual, functional input, and visibility gate on heart and vascular. Appendix Table 4 reports the corresponding edit errors.

Complementary appearance branches. Removing the learned residual reduces PSNR in all eight columns; removing the physical lookup causes larger losses, including 2.57 and 2.65 dB on composition. These results support combining a direct response to the authored edit with learned corrections to reproduce the reference images.

Functional input. Replacing the functional code with preset embeddings assigns an unseen TF the embedding of its nearest training preset. This variant is competitive on nearby vascular edits, but

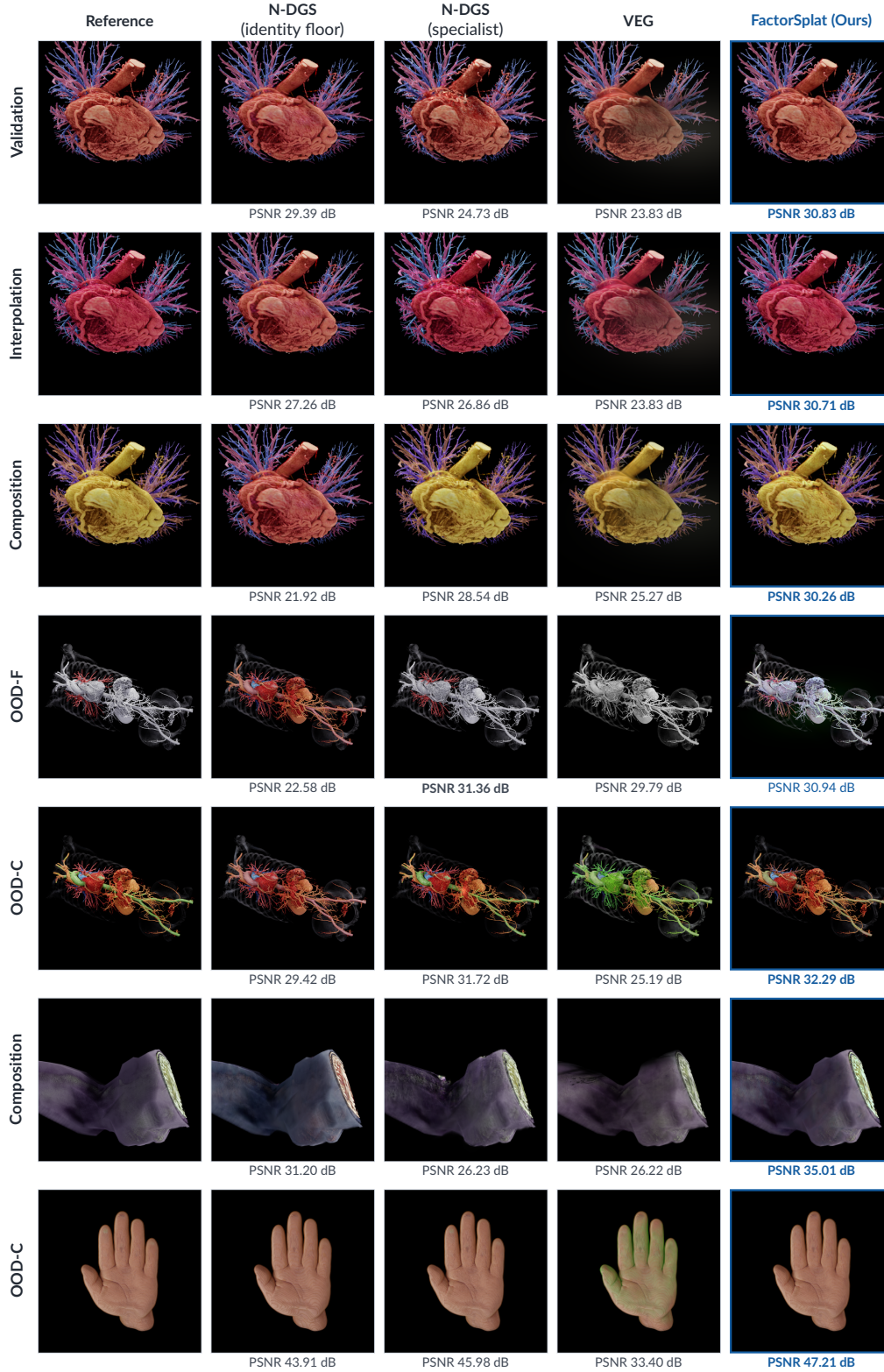


Figure 3: **Held-out CT and MR examples.** Heart (rows 1–3), vascular (4–5), nose (6), and hand (7). Specialists are trained separately on each target TF. Full-frame PSNR; bold marks each row’s best.

Table 2: **Two-scene ablations** ($r=8$). PSNR $\uparrow$, mean rank, and worst gap over eight comparisons.

Variant	heart				vascular				robustness	
	val	int.	cmp	OOD	val	int.	cmp	OOD	rank $\downarrow$	gap (dB) $\downarrow$
FactorSplat (Full)	30.79	29.33	28.68	23.77	34.20	33.68	32.59	30.88	1.69	0.48
– exact gate	30.79	29.33	28.68	23.45	34.20	33.68	32.59	30.36	1.94	0.52
– learned residual	30.16	28.69	27.61	23.22	33.70	32.55	31.92	29.28	3.62	1.60
– physical lookup	29.77	27.48	26.11	20.30	33.31	32.13	29.94	29.18	4.88	3.47
– functional code	30.76	28.38	28.42	22.75	34.30	33.68	33.07	28.04	2.88	2.84

loses 1.02 and 2.84 dB on heart and vascular OOD. The functional input is most useful when the requested curve differs substantially from the training presets.

Visibility control. Removing the analytic gate leaves validation, interpolation, and composition unchanged, while reducing OOD PSNR by 0.32 and 0.52 dB. OOD delta error rises from 9.64 to 10.87 on heart and from 4.54 to 5.01 on vascular ($\times 10^{-2}$). The gate benefits material removal.

4.4 COMPUTATIONAL COST

Rendering. FactorSplat renders faster than VEG, averaging 524 FPS versus 144 FPS across seven scans on B200 GPUs at 1600^2 (Table 1). Its cached fast renderer separates appearance updates from view navigation: conditioning is evaluated once per TF switch, averaging 1.17 ms; cached attributes are reused during camera motion. Both methods support new TFs without refitting. For FactorSplat, FPS uses the fast renderer and PSNR the quality renderer; Appendix E details timing.

Training. FactorSplat also trains faster than VEG: mean training time is 9.8 versus 43.8 minutes per scan for 30k iterations on a B200. One FactorSplat checkpoint supports multiple appearances, whereas N-DGS specialists must be trained separately for every target TF. Across the 17 evaluated TFs, their estimated total training cost averages 88.1 minutes per scan, computed as $17 \times$ one fixed preset’s training time. Sharing geometry and conditioning appearance therefore avoids the repeated optimization required by per-TF specialists.

5 DISCUSSION

Implications for TF authoring. Function-valued conditioning lets users adjust region-specific color and opacity without selecting a different checkpoint. Interpolation and composition probe two complementary aspects of this workflow: adjusting familiar controls and combining edits observed separately during training. The results support both forms of exploration with a shared proxy. They also suggest a useful principle for training-bank design: include distinct appearance operations and expose the materials they affect, rather than simply adding visually similar presets. TF diversity supplies appearance supervision, while coverage-aware pruning retains primitives needed to express those changes. A training bank that combines both types of coverage could extend the range of edits supported by one representation.

Local control and global appearance. Local conditioning ties each primitive’s response to the materials it represents, making authored edits composable within a shared Gaussian field. This locality also limits the response to global illumination changes: edits outside a primitive’s support leave its code unchanged, even when they alter its lighting. Contextual conditioning could address this limitation. Persistent descriptors can also misalign as centers move, and visibility gating removes whole Gaussians rather than clipping at voxel boundaries.

Feed-forward proxy generation. FactorSplat learns one proxy per scan. Feed-forward prediction of shared geometry and conditional appearance factors, building on Render-FM (Gao et al., 2026), could reduce this construction cost while retaining inference-time appearance control.

6 CONCLUSION

FactorSplat combines local TF lookup and a low-rank appearance residual to render unseen region-specific transfer functions with shared Gaussian geometry. Across seven CT and MR scans, it improves mean reconstruction and edit fidelity over region-aware VEG while reducing training time and increasing rendering throughput. The ablations support combining authored optical changes with learned corrections. By decoupling appearance updates from camera motion, FactorSplat supports interactive color and opacity control within one proxy instead of separate models for each preset. Generalizing to a broader range of unseen edit operations remains an important next step.

AI USE STATEMENT

Generative AI tools assisted with method development, experimental design, implementation, result interpretation, and manuscript revision. The authors made the final research decisions and take responsibility for the content.

REPRODUCIBILITY STATEMENT

Sections 3 and 4 specify the model, training procedure, TF and camera splits, baselines, and metrics. The appendix provides the cached evaluation procedure, VEG adaptation, complete per-scan results, ablation edit metrics, and timing protocol. The repository retains experiment scripts, TF manifests, per-view metrics, and figure-source records.

ETHICS STATEMENT

FactorSplat is a research visualization method evaluated on rendered CT and MR images, not a clinically validated diagnostic system. Approximation errors and TF edits can change anatomical visibility. Clinical use would require task-specific validation, expert oversight, and access to the original scan. Medical volumes, masks, and derived artifacts remain subject to source permissions, privacy requirements, and applicable licenses.

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A EFFICIENT TRANSFER-FUNCTION EVALUATION

Appearance scales. All scans use lookup scales $\lambda_L^c = 1$ and $\lambda_L^a = 4$, and residual scales $\lambda_R^c = 0.25$ and $\lambda_R^a = 4$. The lookup gains are learned, while these output scales remain fixed.

Cached transfer-function evaluation. At each TF switch, the shared encoder evaluates the $M \times B$ region–intensity entries once, where M is the number of region-specific transfer curves and $B = 64$ is the number of intensity bins per curve. Each Gaussian aggregates the resulting r -dimensional embeddings over its persistent local support to obtain $\mathbf{z}_{i,T}$. The physical lookup and low-rank residual then update its DC color and opacity. These attributes are cached across camera changes, avoiding repeated TF encoding during view navigation.

Representation size. The conditional model adds $4Nr$ per-Gaussian factors, $H(4 + r)$ shared encoder weights, and four lookup gains. Its persistent descriptors contain at most NK weighted region–intensity samples. The representation therefore scales linearly with the number of primitives, independently of the number of TFs.

B REGION-AWARE VEG BASELINE

We build on the official VEG code (Dyken et al., 2026). To support the multi-region condition $\mathbf{R}_T(m, h)$, our extension assigns each Gaussian a fixed categorical region m_i and one learnable scalar v_i . We retain the released initializer: the input point cloud supplies its bounding box and point count, while positions, scalars, and normals follow VEG’s native random draws. Each primitive receives its nearest source seed’s density-weighted majority region. The authored curves are resampled to VEG’s native $B_V = 100$ intensity knots, giving $\mathbf{R}_T^{(B_V)}$. Its within-region coordinate is $h_i = (B_V - 1) \text{sigmoid}(v_i)$. Linear interpolation is performed only between adjacent intensity bins of the same region:

$$(\mathbf{c}_i(T), a_i(T)) = \text{lerp}\left(\mathbf{R}_T^{(B_V)}(m_i, \lfloor h_i \rfloor), \mathbf{R}_T^{(B_V)}(m_i, \lceil h_i \rceil), h_i - \lfloor h_i \rfloor\right). \quad (14)$$

The queried color enters VEG’s Blinn–Phong shading with learned normals, ambient, diffuse, and specular factors, and shininess. Opacity is $w_i a_i(T)$, where w_i is VEG’s independent sigmoid-activated weight. The authored region-visibility flag is included in TF alpha. Region labels remain fixed under motion and follow the actual parent indices during native cloning and splitting. We retain the upstream optimizer, image losses, regularizers, and weight-based pruning, with scalar and opacity-weight learning rates 2.5×10^{-4} and 2.5×10^{-2} , respectively. Training uses 24 authored presets for 30,000 iterations in place of VEG’s synthetic opacity bank, with no primitive-count cap.

Floating-point and native-quantized evaluation. Both rows use the same trained checkpoint. The float row evaluates its uncompressed parameters. The native-quantized row follows the released deployment path: 256-center codebooks encode opacity, scale, rotation, normal, scalar, diffuse, and ambient attributes; codebook centers and the remaining continuous attributes use half precision. We evaluate decoded parameters, with region assignments preserved losslessly. Quality and timing are measured separately for each representation. Quantization lowers mean PSNR by 0.15/0.12/0.10/0.05 dB on validation/interpolation/composition/OOD.

Numerical stability. We fix a numerical bug in the released VEG implementation that produces non-finite gradients and interrupts training. The native specular term $|\mathbf{n}_i^\top \mathbf{h}|^{s_i}$ has a singular derivative at zero when $0 < s_i < 1$, where $\mathbf{h}$ is the half-vector and s_i is shininess. We mask exact-zero inputs before exponentiation and assign zero gradients at those inputs, preserving forward values and gradients elsewhere. With this correction, VEG completes training on all seven scans.

C PER-SCAN RESULTS

Table 3 reports each scan separately and the unweighted seven-scan mean. Each split averages its held-out TFs, with five edits contributing to OOD. FactorSplat leads VEG in PSNR across all splits on vascular, intestine, lower, nose, and hand; VEG has lower edit error throughout heart and kneejoint.

Table 3: **Per-scan comparison.** Changed-region Δ_{TF} ($\times 10^{-2}$) and PSNR; OOD averages five edits and (h) is the unweighted mean. Bold marks column best.

Model	$\Delta_{TF} \downarrow (\times 10^{-2})$				PSNR $\uparrow$			
	val	interp	comp	OOD	val	interp	comp	OOD
(a) heart (CT, 12 labels)								
N-DGS (identity floor)	3.74	6.76	10.28	19.92	28.31	25.18	21.49	17.77
N-DGS (specialist)	—	—	—	—	28.82	29.06	29.21	30.47
VEG	1.22	2.26	3.03	5.93	28.79	28.21	27.80	25.70
VEG (native-quantized)	1.22	2.26	3.03	5.95	28.69	28.13	27.73	25.66
FactorSplat (Ours)	1.31	2.63	3.14	9.64	30.79	29.33	28.68	23.77
(b) vascular (CT, 15 labels)								
N-DGS (identity floor)	3.93	7.53	10.63	13.81	32.40	29.36	26.83	25.15
N-DGS (specialist)	—	—	—	—	32.27	32.45	32.59	33.16
VEG	1.61	3.58	4.37	10.02	31.95	30.91	30.31	25.68
VEG (native-quantized)	1.62	3.59	4.39	10.03	31.47	30.55	30.03	25.56
FactorSplat (Ours)	1.60	2.48	3.59	4.54	34.20	33.68	32.59	30.88
(c) intestine (CT, 17 labels)								
N-DGS (identity floor)	3.40	6.61	8.34	14.97	30.79	26.92	25.63	21.58
N-DGS (specialist)	—	—	—	—	29.27	29.06	29.35	30.07
VEG	1.73	3.06	3.35	10.25	31.61	29.81	29.75	24.20
VEG (native-quantized)	1.74	3.08	3.36	10.24	31.56	29.76	29.69	24.18
FactorSplat (Ours)	1.30	2.08	2.85	9.95	32.53	31.60	30.47	25.53
(d) lower (CT, 7 labels)								
N-DGS (identity floor)	3.69	6.95	10.10	17.33	32.68	29.07	27.00	24.10
N-DGS (specialist)	—	—	—	—	32.10	32.38	32.86	33.66
VEG	1.33	2.42	3.03	8.26	32.93	32.38	32.45	28.40
VEG (native-quantized)	1.33	2.42	3.04	8.27	32.88	32.34	32.41	28.38
FactorSplat (Ours)	1.29	2.16	3.08	4.61	34.30	33.55	32.66	30.87
(e) knee joint (CT, 3 labels)								
N-DGS (identity floor)	3.24	4.58	5.47	22.67	31.37	28.87	26.54	16.95
N-DGS (specialist)	—	—	—	—	31.05	31.13	31.44	32.36
VEG	1.14	1.42	1.47	11.30	33.72	33.21	32.91	22.85
VEG (native-quantized)	1.14	1.43	1.48	11.36	33.63	33.12	32.85	22.80
FactorSplat (Ours)	1.31	1.47	1.73	11.74	33.59	33.29	32.24	21.87
(f) nose (MR, 3 labels)								
N-DGS (identity floor)	2.96	4.14	4.48	14.38	36.60	35.02	33.04	25.06
N-DGS (specialist)	—	—	—	—	34.94	34.80	34.91	36.85
VEG	1.48	2.08	1.98	6.14	36.69	36.19	35.54	29.35
VEG (native-quantized)	1.48	2.08	1.98	6.13	36.51	36.02	35.40	29.30
FactorSplat (Ours)	1.33	1.71	1.60	5.50	38.27	37.86	37.49	30.94
(g) hand (MR, 9 labels)								
N-DGS (identity floor)	2.80	4.35	6.48	7.80	36.55	36.06	29.30	30.65
N-DGS (specialist)	—	—	—	—	39.12	39.29	39.23	39.16
VEG	0.90	1.71	1.20	4.18	40.94	40.85	40.09	33.13
VEG (native-quantized)	0.90	1.71	1.20	4.18	40.83	40.75	40.00	33.11
FactorSplat (Ours)	0.85	1.53	0.96	3.79	43.09	42.87	42.40	35.32
(h) average								
N-DGS (identity floor)	3.39	5.85	7.97	15.84	32.67	30.07	27.12	23.04
N-DGS (specialist)	—	—	—	—	32.51	32.59	32.80	33.67
VEG	1.34	2.36	2.63	8.01	33.80	33.08	32.69	27.05
VEG (native-quantized)	1.35	2.37	2.64	8.02	33.65	32.95	32.59	27.00
FactorSplat (Ours)	1.29	2.01	2.42	7.11	35.25	34.60	33.79	28.45

D ABLATION EDIT FIDELITY

Table 4 complements the reconstruction metrics in Table 2 with changed-region Δ_{TF} for the same variants. Removing either appearance branch increases edit error in all eight columns. Preset embeddings remain competitive on nearby vascular conditions, whereas the functional code improves OOD edit fidelity on both scans. The visibility gate affects only presets that explicitly remove material. Unchanged-region leak spans $0.12\text{--}0.43 \times 10^{-2}$ across these variants. In Table 2, mean rank averages ranks across eight PSNR columns; worst gap is the maximum deficit from a column’s best score.

Table 4: **Ablation edit fidelity.** Changed-region Δ_{TF} ($\times 10^{-2}$; lower is better).

Variant	heart				vascular			
	val	int.	cmp	OOD	val	int.	cmp	OOD
FactorSplat (Full)	1.31	2.63	3.14	9.64	1.60	2.48	3.59	4.54
– exact gate	1.31	2.63	3.14	10.87	1.60	2.48	3.59	5.01
– learned residual	1.39	2.87	3.62	10.56	1.67	3.40	4.26	6.83
– physical lookup	2.86	5.27	6.59	14.71	2.95	4.67	6.98	7.77
– functional code	1.35	3.46	3.17	10.92	1.50	2.43	3.17	7.85

E RUNTIME MEASUREMENTS

Rendering protocol. Table 5 compares the two methods across seven scans on B200 GPUs at 1965 MHz. Each method renders the same eight validation views with five warm-ups and 20 CUDA-event measurements per view. FPS is the inverse of mean frame time; Table 1 averages it equally across scans within each modality. N-DGS references use the same fast rasterizer and timing protocol, with the appropriate specialist selected before timing. Measurements include VEG’s native TF lookup, shading, rasterization, and auxiliary outputs, and FactorSplat’s cached fast rendering; data loading, image saving, and scoring are excluded. PSNR is measured separately with the quality renderer over all 680 held-out views (17 TFs at 40 cameras each). Fast and quality rendering can produce different images, so the table compares deployment throughput alongside reconstruction quality. Primitive counts provide context for the rendering cost.

Table 5: **B200 deployment at 1600².** Matched views; FactorSplat FPS uses cached fast rendering. PSNR uses quality rendering.

Scene	N (k)		PSNR $\uparrow$		FPS $\uparrow$		Speedup
	Ours	VEG	Ours	VEG	Ours	VEG	
heart	278.6	146.3	27.89	27.51	477	109	4.36 $\times$
vascular	235.7	148.1	32.72	29.47	525	181	2.90 $\times$
intestine	281.8	208.9	29.77	28.57	494	122	4.06 $\times$
lower	250.5	127.0	32.73	31.36	531	155	3.42 $\times$
kneejoint	80.4	82.7	29.75	30.21	515	103	5.03 $\times$
nose	150.2	111.9	35.84	34.14	564	146	3.87 $\times$
hand	135.1	54.7	40.59	38.42	560	193	2.89 $\times$

Transfer-function updates. TF-update latency measures appearance-cache construction separately, with five warm-ups and 20 synchronized wall-time measurements per view. This cost is incurred once per TF switch, not during fixed-TF view navigation.

F ADDITIONAL QUALITATIVE COMPARISONS

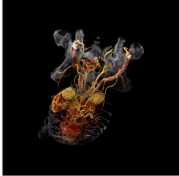
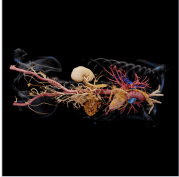
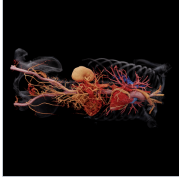
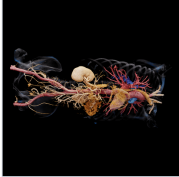
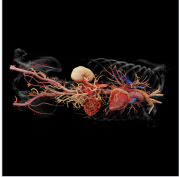
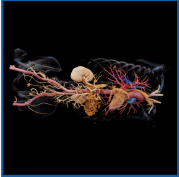
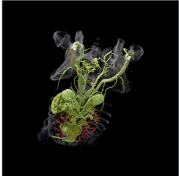
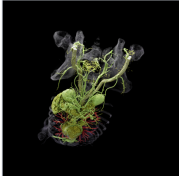
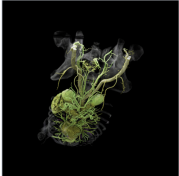
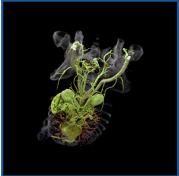
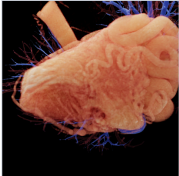
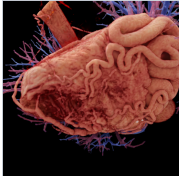
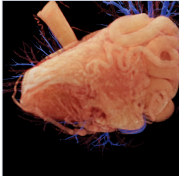
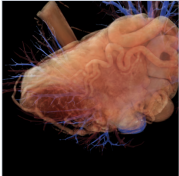
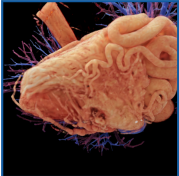
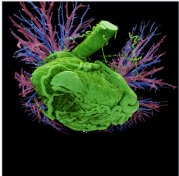
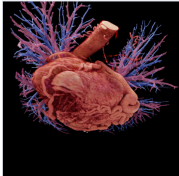
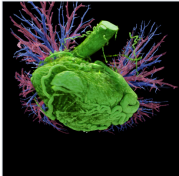
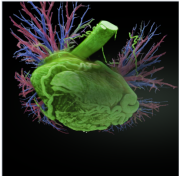
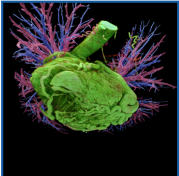
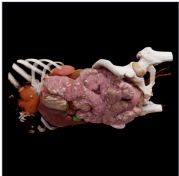
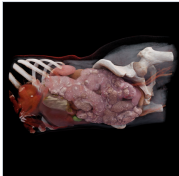
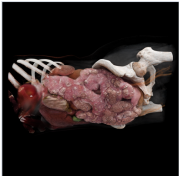
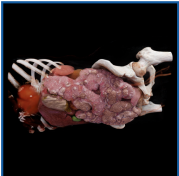
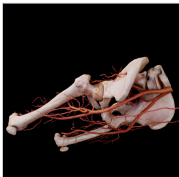
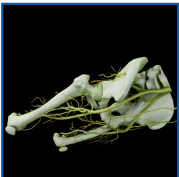
	Reference	N-DGS (identity floor)	N-DGS (specialist)	VEG	FactorSplat (Ours)
Validation		 PSNR 32.80 dB	 PSNR 28.83 dB	 PSNR 31.14 dB	 PSNR 35.73 dB
Interpolation		 PSNR 27.23 dB	 PSNR 36.45 dB	 PSNR 29.92 dB	 PSNR 34.93 dB
Composition		 PSNR 27.15 dB	 PSNR 33.29 dB	 PSNR 30.45 dB	 PSNR 34.50 dB
OOD-F		 PSNR 18.35 dB	 PSNR 31.50 dB	 PSNR 21.33 dB	 PSNR 23.92 dB
OOD-C		 PSNR 18.18 dB	 PSNR 27.61 dB	 PSNR 22.90 dB	 PSNR 27.68 dB
Interpolation		 PSNR 22.00 dB	 PSNR 34.49 dB	 PSNR 26.43 dB	 PSNR 32.36 dB
Composition		 PSNR 24.91 dB	 PSNR 25.89 dB	 PSNR 19.64 dB	 PSNR 32.33 dB

Figure 4: **Additional held-out CT examples.** Vascular (rows 1–3), heart (4–5), intestine (6), and lower body (7). Full-frame PSNR; bold marks each row’s best.